

Question 1: What is the primary goal of data science?

- ☐ Predicting future events with 100% accuracy
- ☒ Learning about the world from data using computation
- ☐ Creating complex algorithms for data manipulation
- ☐ Generating random datasets for experimentation

Question 2: Which of the following represent the stages of the Data Science Life Cycle?

- ☐ Data Mining, Data Validation, Model Deployment, Model Evaluation
- ☒ Problem Formulation, Data Acquisition, EDA, Inference and Prediction
- ☐ Data Visualization, Data Cleansing, Model Training, Model Testing
- ☐ Data Storage, Data Retrieval, Data Preprocessing, Model Interpretation

Question 3: Which components are typically involved in the design of a robotic agent?

- ☐ Sensors, Environment, Programming Languages
- ☐ Cameras, Motors, Software Libraries
- ☒ Sensors, Actuators, Environment
- ☐ Hardware, Software, Network Connectivity

Question 4: Which of the following accurately describes Python programming language?

- ☐ Compiled, specialized, low-level syntax with strict semantics
- ☐ Interpreted, domain-specific, high-level syntax with dynamic semantics
- ☒ Interpreted, general-purpose, high-level language with easy syntax and dynamic semantics
- ☐ Compiled, general-purpose, low-level syntax with strict semantics

Question 5: Which of the following statements accurately compare NumPy arrays with simple Python lists?

- ☐ NumPy arrays are slower and less versatile than Python lists.
- ☐ Python lists are more efficient in handling N-dimensional arrays compared to NumPy.
- ☒ NumPy arrays are the fundamental package for scientific computing, offering speed and versatility not found in Python lists.
- ☐ Python lists provide better vectorization, indexing, and broadcasting capabilities compared to NumPy.

Question 6: Which of the following statements accurately differentiate between NumPy and Pandas libraries?

- ☐ NumPy is designed for working with tabular or heterogeneous data, while Pandas is best suited for homogenous numerical data.
- ☐ Pandas is derived from Panel Data and is primarily used for numerical computations, while NumPy is a data analysis library.
- ☒ Pandas adopts many coding idioms from NumPy, but the biggest difference lies in their suitability: Pandas for tabular or heterogeneous data, and NumPy for homogenous numerical data.
- ☐ NumPy offers fast and convenient data cleaning and analysis in Python, while Pandas is designed for working with specialized numerical arrays

Question 7: What are the fundamental data structures in the Pandas library?

- ☐ Series, DataFrame, Vector
- ☐ Array, DataFrame, Indexer
- ☐ Series, Data Table, Index
- ☒ Series, DataFrame, Index

Question 8: Which statements accurately describe slicing methods in Pandas DataFrame?

- ☒ The ``loc`` method selects items by label, where the first argument represents rows and the second argument represents columns. The ``iloc`` method selects items by number, following the same order of arguments as ``loc``. The square brackets ``[]`` can only take one argument, which may be a slice of row numbers, a list of column labels, or a single column label.
- ☐ The ``iloc`` method selects items by label, while the ``loc`` method selects items by number. The square brackets ``[]`` can take two arguments: a slice of row numbers and a list of column labels.
- ☐ The ``loc`` method selects items by number, while the ``iloc`` method selects items by label. The square brackets ``[]`` can only take one argument, which must be a list of column labels.
- ☐ The ``loc`` method selects items by label, where the first argument represents columns and the second argument represents rows. The ``iloc`` method selects items by number, following the same order of arguments as ``loc``. The square brackets ``[]`` can only take one argument, which may be a slice of row numbers, a list of column labels, or a single column label.